



Lava Network: Building the future of blockchain with scalable cloud infrastructure

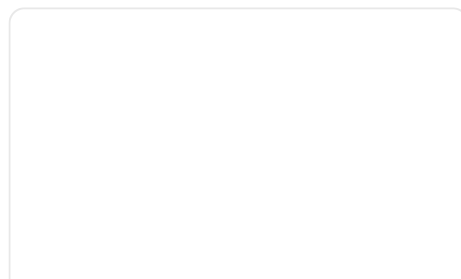
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GOOGLE CLOUD RESULTS

- ✓ Grows traffic by 60x while keeping cloud costs under control with [Google Kubernetes Engine](#)
- ✓ Offers reliable blockchain access to users and developers with 99.999% availability
- ✓ 90 billion RPC requests served to date with low latency, reliable [Google Cloud infrastructure](#)

Migrating to Google Cloud enabled Lava Network to scale its traffic 60x, while working with the Google Cloud Web3 team to grow the blockchain ecosystem.

Blockchain is the foundation on which the [vision of Web3](#) is built: a decentralized internet, owned by its users. Originally conceived to make



cryptocurrency a reality, its inherent security and efficiency mean it's now used for everything from real estate deals and healthcare records to supply chain management and transparent voting systems.



For Web3 to become truly mainstream, it needs infrastructure that can scale. New blockchains are launched every year to support a growing number of users and applications, but for developers to build their decentralized applications (dApps) on those blockchains, they need reliable access to the blockchains' data.

The RPC nodes that provide this access can be costly for the blockchains to run and often provide slow and unreliable connections. Recognizing that this was creating a bottleneck for multichain dApp development, the [Lava Network](#) launched Lava, using aggregated pools like RPC node runners to provide easy, reliable, and affordable access to developers to build on blockchain.

"Blockchain enables a fairer, more equal, and user-owned internet," says Ethan Luc, GTM and Community at Lava Network. "To have this internet built on blockchains, you need an effective data layer. That's what Lava provides—we're making it easy for users and builders to interact with and build blockchain apps."



The need for cloud infrastructure to support rapid growth

Lava was soon supporting hundreds of node providers on dozens of blockchains when it began to run into scalability challenges of its own. The more traffic it handled, the more its cloud bill with its

third-party provider rose. As Lava prepared to launch its new Magma points program to incentivize users to interact with the Lava network in exchange for points, the company envisaged a major uptick in users. Lava knew it would need reliable, affordable infrastructure to support its next stage of growth.

“Our developers were really impressed with the ease of use of Google Cloud, and we liked that it gave us the ability to understand what we were actually paying for. It also provided everything we needed in one place, including native support for the Web3 domain.”

Nimrod Teich

VP R&D, Magma Devs, Contributors to Lava Network

As Lava searched for a more cost-effective and dependable cloud provider, its developers spoke highly of their prior experiences using [Google Cloud](#).



A cost-efficient way to increase traffic

Lava migrated to Google Cloud at the start of 2024, and began running its systems on [Google Kubernetes Engine \(GKE\)](#). When it launched its points program soon afterward, Lava’s traffic volume increased by 60x in the space of one month. With GKE’s autoscaling feature, Lava could manage that increase by automatically scaling up its compute power to handle the high load, before scaling it back down during periods of lower demand to ensure it was only paying for the resources it used.

“Autoscaling with GKE was a game changer that allowed us to scale up our operations to support more traffic and more requests per day,” says Nimrod Teich, VP R&D, Magma Devs, part of the Lava Network. “We also worked closely with Google Cloud to understand what resources we were likely to use, and agreed a committed-use discount to keep our costs under control.”

Lava further improved the cost-efficiency of its cloud spend by using [Cloud Armor](#) to identify and filter out bot traffic to ensure it wasn't paying for unwanted requests. "With a 60x increase in traffic, it was very important not to have exponential growth in expenditure, so we used a lot of different Google Cloud solutions, such as Cloud Armor, to mitigate that risk," Teich explains.



Delivering a fast, reliable blockchain experience

Running on scalable Google Cloud infrastructure has allowed Lava to deliver a reliable service, with 99.999% availability to developers and end users. "We have had 2.5 million unique users in the last 30 days and we support 150,000 users daily. All of that scaling is in large part down to our migration to Google cloud," says Luc. For end users, a reliable service means they can access the dApps and services they want, such as checking their crypto balance or buying an NFT, without any interruption.

Low latency is another contributing factor. With a network of [data centers](#) spanning 40 regions, Google Cloud allows Lava to deploy kubernetes clusters in regions all over the world to give developers and end users fast access to blockchain data. "When latency improves, so does the service for everyone in the blockchain industry," explains Teich. "Latency

"If a blockchain's RPC doesn't work well, no one's going to build on it. With the help of Google Cloud, we're helping smaller, newer blockchains get the same level of infrastructure as the bigger blockchains and attract users and builders to build on their chains."

Ethan Luc

GTM and Community,
Lava Network

matters, because say you are making a crypto transaction, the system might time out before the transaction completes, which can be very frustrating when money is involved.”

Low latency and high availability benefit the RPC node providers operating on the Lava network too. Node providers are incentivized to serve as much traffic as possible, and the way they attract traffic is by offering a reliable service, which Lava provides with Google Cloud. With more than 400 RPC providers now operating on Lava at peak, the network has served a total of 90+ billion RPC requests to date.

The blockchains benefit, in turn, by attracting more developers to build on their network. “If a blockchain’s RPC doesn’t work well, no one’s going to build on it,” explains Luc. “With the help of Google Cloud, we’re helping smaller, newer blockchains get the same level of infrastructure as the bigger blockchains and attract users and builders to build on their chains.”

Lava benefits from the high availability of other Google Cloud solutions too. With a customized [usage dashboard](#), Lava allows dApp developers to explore the performance data of various blockchains. This data is all stored in [Cloud SQL](#). “Some managed SQL instances don’t work well,” says Teich. “They quickly max out your resources, or don’t optimize your queries properly. Cloud SQL just works. We don’t have to do too much. We don’t have to pay much to receive the service we want. And again, there’s zero downtime.”



Building together toward a multi-chain world

“We believe that we are heading toward a multi-

Lava is also making use of Google Cloud’s own extensive [Web3 infrastructure](#), using [Blockchain Node Engine](#) to run nodes of its own to offer developers fast and reliable

chain world with many blockchains. With Google Cloud Lava Network can provide scalable, available, and fast access to any blockchain for millions of daily users—and help to make that vision a reality.”

Ethan Luc
GTM and Community,
Lava Network

access to the Ethereum blockchain. Now, Lava is working together with Google Cloud to enable people who are already using Blockchain Node Engine on their own nodes to use any excess capacity in those nodes to offer developers access to Lava too. “Anyone running a node never uses the entire compute resources—a lot of it goes to waste,” explains Yuval Binder, BD & Customer Success at Magma Devs, Contributors to the Lava Network. “So if you can use those resources to make extra income and cover some of your running costs, that would be very appealing to enterprises and institutions currently using Google to run their own nodes.”

With these projects and a series of further points programs, partnerships, and regional expansions coming up, Lava is continuing its steep growth curve as it attracts more and more traffic to its network. “We believe that we are heading toward a multi-chain world with many blockchains,” explains Luc. “With Google Cloud, Lava Network can provide scalable, available, and fast access to any blockchain for millions of daily users—and help to make that vision a reality.”

Lava Network is the most performant and reliable platform for blockchain data across 40+ networks. By aggregating RPC and data providers, Lava can automatically load balance and has handled 80bn+ requests with 99.999% uptime for both large and newer chains. Lava abstracts away the complexity of multi-chain development. Developers get a free endpoint for every supported chain, with generous rate limits. Lava provides access to any data, on every blockchain.

Industry: Technology

Location: Israel

Products: [Google Kubernetes Engine \(GKE\)](#), [Blockchain Node Engine](#), [Cloud Armor](#), [Cloud SQL](#), [Pub/Sub](#), [Web3 infrastructure](#)

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